Climate Action: Plan, Measure, Improve

I can explain how an action reduces climate impact, distinguish reducing causes from coping with effects, and plan evidence to check whether it worked.

Mitigation reduces a cause adaptation reduces harm from an effect

A sensible action still needs a mechanism and evidence

Measure before and after in the same way before claiming success

Open the original interactive lesson

What success looks like

- I can distinguish mitigation from adaptation.
- I can connect an action to a scientific mechanism.
- I can choose a baseline and repeated measure before making a claim.

Retrieve and reconnect

- State how switching off an unused light could reduce emissions.
- Explain one difference between mitigation and adaptation.
- Name one measure needed before an action starts.
- Explain why a poster alone is not evidence of impact.

Which plan reduces a cause, and which plan helps people cope with an effect?

Think first. Point, say, sign or write your choice. Give one reason.

A. Switch off lights in empty rooms

B. Provide safe shade during high heat

C. Put up a poster and take no measurements

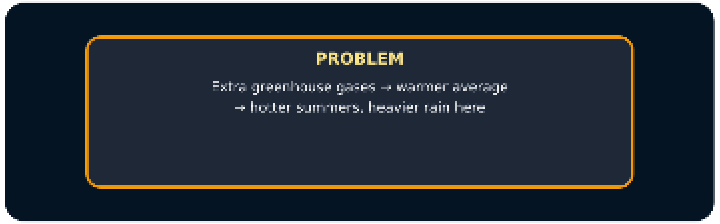
Look closely · what is the evidence?

PROBLEM · name it

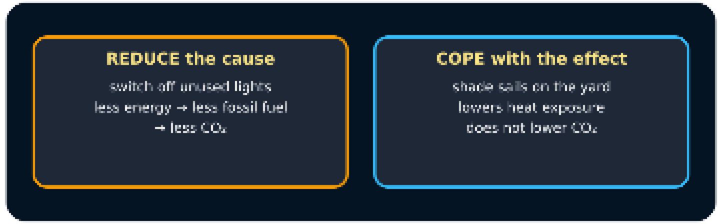
ACTION · explain the mechanism

MEASURE · compare and review

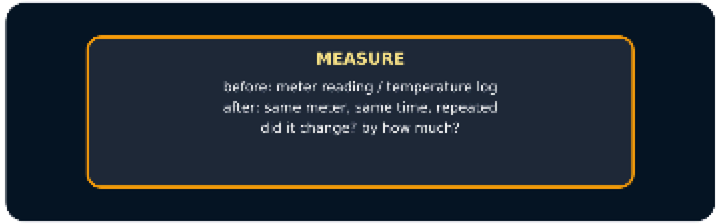
Look closely · what is the evidence?



Problem card



Two kinds of action



Measure card

Words we will use

- mitigation — reduce a cause
- adaptation — cope with an effect
- baseline — what happens before

Watch the model

- Problem: lights are sometimes left on in empty rooms, using electricity unnecessarily.
- Action and mechanism: switch them off safely; where fossil fuels supply energy, lower demand can mean fewer emissions.
- Contrast: safe shade lowers heat exposure. It is useful adaptation, but it does not directly reduce the cause.

Which plan can check whether the lighting action worked?

A. Record unused light-on minutes for five days before and five days after, in the same way.

B. Put up a poster, then state that the problem is solved.

C. Ask one person whether the climate feels better.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Baseline: record what happens before the change. Prediction: state what should change and why.
- Action: make only a teacher-approved safe change. Measure repeatedly using the same rule.
- Review: compare the evidence and keep 'not yet known' if the evidence is too weak.

Find and repair the misconception

Wasted-light time fell, so this one action fixed climate change.

A. Keep it: one local result proves the global outcome.

B. Repair it: the sample supports less wasted-light time here, not a global climate claim.

C. Erase the result because local evidence is never useful.

Before/After Evidence Lab

Sort each card as Mitigation, Adaptation or Evidence check. Use the fictional data only to practise reasoning.

- Name the problem
- Classify the action
- Compare the baseline
- State a limited conclusion

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- MITIGATION
- ADAPTATION
- EVIDENCE CHECK

State only what the sample supports, then name one thing it cannot show.

Your independent mission

Choose one approved school scenario and create an action, mechanism and measurement plan before making any success claim.

Record: A safe action-to-mechanism chain, baseline, repeated measure, comparable condition and one limitation.

Choose your support and challenge

- Supported: Select labelled cards for Action, How it helps and What we measure.
- Standard: Complete the full plan with a baseline, prediction, repeated measure and one controlled condition.
- Stretch: Compare mitigation with adaptation and identify one confounding factor or evidence limit.

Show what you know

Explain one action-to-mechanism chain.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.